

Problem Submissions Discussions

Height of Binary Search Tree

Given the root of a binary search tree, return the height of the tree. Height is the number of nodes along the longest path from root to leaf.

Example

Input

```
n = 7
values = [4, 2, 6, 1, 3, 5, 7]
leftChild = [1, 3, 5, -1, -1, -1, -1]
rightChild = [2, 4, 6, -1, -1, -1, -1]
```

Output

3

Explanation

The tree is perfectly balanced with three levels:
- Level 1: Node 4 (root)
- Level 2: Nodes 2 and 6
- Level 3: Leaves 1, 3, 5, 7 The longest path from root to any leaf has 3 nodes, so the height is 3.

Input Format

The first line contains n : integer denoting number of nodes in the binary search tree.
The next n lines contains individual elements of the array, $values[i]$ is the integer value stored in the i^{th} node.
The next line contains an integer m denoting length of $leftChild$.
The next m lines contains the elements of $leftChild$. $leftChild[i]$ is the index of the left child of node i , or -1 if none.
The next line contains an integer p denoting length of $rightChild$.
The next p lines contains the elements of $rightChild$. $rightChild[i]$ is the index of the right child of node i , or -1 if none.

Constraints

$0 \leq n \leq 1000$
 $values.length == n$
 $leftChild.length == n$
 $rightChild.length == n$
For all $0 \leq i < n$: $-10^9 \leq values[i] \leq 10^9$
All $values[i]$ are unique
For all $0 \leq i < n$: $leftChild[i] == -1$ or $(0 \leq leftChild[i] < n)$
For all $0 \leq i < n$: $rightChild[i] == -1$ or $(0 \leq rightChild[i] < n)$
For all $0 \leq i < n$: if $leftChild[i] != -1$ then $values[leftChild[i]] < values[i]$
For all $0 \leq i < n$: if $rightChild[i] != -1$ then $values[rightChild[i]] > values[i]$
The set of edges defined by $(i, leftChild[i])$ and $(i, rightChild[i])$ forms a single connected acyclic graph (a tree)

Output Format

Return a single integer representing the height of the tree: the number of nodes along the longest path from the root to any leaf. If $n = 0$, return 0.

Sample Input 0

```
1
10
1
-1
1
-1
```

Sample Output 0

1

Sample Input 1

```
2
5
3
2
1
-1
2
-1
-1
```

Sample Output 1

2

Language Python 3

```
1 > #!/bin/python3 --
10
11 #
12 # Complete the 'getBinarySearchTreeHeight' function below.
13 #
14 # The function is expected to return an INTEGER.
15 # The function accepts following parameters:
16 # 1. INTEGER_ARRAY values
17 # 2. INTEGER_ARRAY leftChild
18 # 3. INTEGER_ARRAY rightChild
19 #
21 def getBinarySearchTreeHeight(values, leftChild, rightChild):
22     # Write your code here
23
24 > if __name__ == '__main__': --
```

Test Results

Ln 10, Col 1 Autocomplete Spaces: 4 Mode: Normal

Run Code

Submit Code